

CSCI 2406: Topic 02

RISC Principles and ARM Architecture

RISC Design Principles, AArch64 Architectural Overview,
Registers and Programmer-Visible State, and Instruction Classes

Computer Organization & Architecture | AArch64 Reference

Topic 2 Outline

Topic Objective

Examine the design principles commonly associated with RISC architectures and show how those principles appear concretely in AArch64.

- **Section 1:** RISC Design Principles
- **Section 2:** AArch64 Architectural Overview
- **Section 3:** Registers and Programmer-Visible State
- **Section 4:** Core AArch64 Instruction Classes
- **Section 5:** Summary and Self-Test

Section 01

RISC Design Principles

What Does RISC Mean?

Reduced Instruction Set Computer

RISC is a design philosophy emphasizing **regularity**, **simple instruction forms**, and a clean separation between computation and memory access.

- RISC does **not** simply mean “a processor with very few instructions.”
- The important idea is regular instruction structure.
- Arithmetic and logical operations are primarily register-based.
- Load and store instructions transfer data between registers and memory.

Foundational RISC Design Principles

Instruction Regularity

- Regular instruction formats.
- Consistent operand fields.
- Predictable instruction alignment.

Register-Centered Computation

- Arithmetic uses registers.
- Memory access uses loads and stores.
- Regularity simplifies decoding.

Implementation Benefit

These characteristics support regular instruction fetch, decode, and pipelined implementation.

The Load/Store Architectural Model

Separation of Memory Access and Computation

In a load/store architecture such as AArch64, arithmetic and logical instructions operate on registers. Memory is accessed using dedicated **load** and **store** instructions.

Data Processing

- Operands are registers.
- Results are written to registers.

```
add x1, x2, x3
```

Memory Access

- Loads read memory into registers.
- Stores write registers to memory.

```
ldr x1, [x0]
```

Load–Compute–Store in AArch64

Suppose the address in x0 identifies a 64-bit value in memory that must be increased by 5.

```
ldr x1, [x0]           // load memory value into x1
```

```
add x1, x1, #5         // compute using register operands
```

```
str x1, [x0]           // store result back to memory
```

Architectural Rule

The add instruction does not directly read or modify a memory operand.

Visualizing Load/Store Separation



Takeaway

Loads and stores connect memory to registers; arithmetic instructions operate on register operands.

RISC and CISC: Different Design Traditions

Characteristic	CISC Tradition	RISC Tradition
Instruction format	Often variable length	Typically more regular
Memory operands	Some instructions combine memory access and computation	Load/store separation
Addressing	Often many addressing forms	Comparatively regular forms
Decode	Can require more complex parsing	Regular fields simplify decode
Example	x86-64	AArch64, RISC-V

RISC and CISC are architectural traditions, not simple measures of modern processor complexity.

Same Operation, Different ISA Style

Suppose a 64-bit memory value must be increased by 5.

x86-64 Style

Some x86-64 instructions permit memory operands directly:

```
add qword ptr [rax], 5
```

One instruction describes both memory access and arithmetic.

AArch64 Load/Store Style

```
ldr x1, [x0]
```

```
add x1, x1, #5
```

```
str x1, [x0]
```

Memory transfer and arithmetic are represented by separate instructions.

Section 02

AArch64 Architectural Overview

ARM and AArch64

ARM Architecture

ARM defines a family of instruction set architectures used in embedded systems, mobile computing, desktop systems, and servers.

- **AArch64** is the 64-bit execution state introduced with ARMv8-A.
- It provides a modern 64-bit programmer-visible register model.
- It follows a load/store design with regular instruction formats.
- Different processors can implement AArch64 using different microarchitectures.

AArch64 and A64 Terminology

AArch64

- The architecture / execution state.
- Defines programmer-visible state.
- Includes registers, memory behavior, and instruction semantics.

A64

- The instruction set encoding.
- Used while executing in AArch64 state.
- A64 instructions are fixed 32-bit words.

Course Terminology

Use **AArch64** by default. Use **A64** only when referring specifically to the instruction set encoding.

Key Characteristics of AArch64

Instruction Model

- Load/store architecture.
- A64 instructions are 32 bits wide.
- Instructions are 4-byte aligned.
- Register fields are regular.

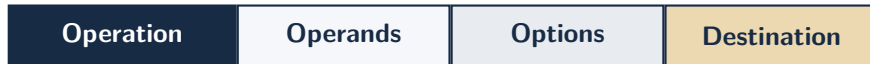
Register Model

- 31 general-purpose registers.
- 64-bit x register views.
- 32-bit w register views.
- Special roles for sp, xzr, PC state, and NZCV.

A64 Instruction Format Characteristics

Fixed-Width Encoding

Every instruction in the A64 instruction set occupies exactly **32 bits**.

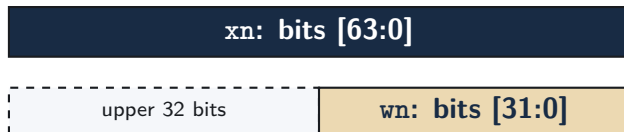


- Fixed width simplifies instruction alignment and initial decode.
- Exact bit-field layouts vary by instruction class.

Detailed machine-code field layouts are covered later in the instruction-encoding topic.

Dual Register Views: X and W

Each general-purpose register can be accessed using either a 64-bit or 32-bit architectural view.



32-Bit Write Rule

Writing to a **wn** destination automatically clears bits [63:32] of the corresponding **xn** register.

Example: 32-Bit Write Clears Upper Bits

Instruction

```
add w3, w1, w2
```

- The operation uses the lower 32-bit views: w1 and w2.
- The 32-bit result is written into w3.
- The upper half of x3 is cleared automatically.

$$x3[63:32] \leftarrow 0$$
$$x3[31:0] \leftarrow \text{result}$$

Section 03

Registers and Programmer-Visible State

AArch64 General-Purpose Registers

Register Set

AArch64 provides **31 general-purpose registers**, named x_0 through x_{30} . Each register is 64 bits wide.

64-Bit View

x_0 – x_{30}

- Each x_n holds 64 bits.
- Used for integer values, addresses, and general computation.

32-Bit View

w_0 – w_{30}

- Each w_n refers to bits [31:0] of x_n .
- Writing w_n clears bits [63:32].

Important

The ISA treats x_0 – x_{30} as general-purpose registers. Software conventions assign additional roles during procedure calls.

The Zero Register

`xzr` / `wzr`

The zero register has two special behaviors: **reads return zero** and **writes are discarded**.

Use as a Zero Operand

```
mov x0, xzr
```

- Reading `xzr` produces the value 0.
- Therefore, `x0` receives zero.

Use as a Discard Destination

```
cmp x1, x2
```

- `cmp` is an alias for `subs xzr, x1, x2`.
- The subtraction result is discarded, while `NZCV` flags are updated.

Key Idea

`xzr` behaves like a permanent zero value when read, and like a “trash destination” when written.

The Stack Pointer

sp

The stack pointer identifies the current stack location and has dedicated architectural semantics.

- Used by stack-related load/store and arithmetic forms.
- Represents a memory address, not a normal scratch register.
- Software conventions impose alignment and usage requirements.
- Stack behavior is studied more deeply with procedures and function calls.

```
stp x29, x30, [sp, #-16]!
```

Encoding Note: Register Number 31

Context-Dependent Meaning

In several A64 instruction formats, register encoding value 31 has special meaning.

Often Means Zero Register

`xzr` / `wzr`

Common in data-processing contexts.

Sometimes Means Stack Pointer

`sp` / `wsp`

Common in stack and memory-addressing contexts.

The instruction definition determines the meaning.

The Program Counter

Program Counter State

The program counter identifies the instruction address associated with the current flow of execution.

- Sequential execution advances to the next instruction.
- Branches and calls redirect execution to another address.
- Unlike AArch32 `r15`, the AArch64 program counter is **not** an ordinary general-purpose register.
- AArch64 arithmetic instructions cannot use `pc` as a normal register operand.

Condition Flags: NZCV

Some AArch64 instructions update four condition flags used by later operations.



- **N**: result has its sign bit set.
- **Z**: result is zero.
- **C**: carry-out for addition; no-borrow for subtraction.
- **V**: signed arithmetic overflow.

Programmer-Visible State: Summary

State	Architectural Role
x0–x30	31 general-purpose 64-bit registers
w0–w30	Lower 32-bit views of corresponding x registers
xzr/wzr	Zero operand; writes are discarded
sp	Stack pointer with dedicated semantics
Program Counter	Identifies control-flow position; not an ordinary GPR
NZCV	Condition flags used by flag-setting and conditional operations

Section 04

Core AArch64 Instruction Classes

Three Core Instruction Classes

For this course, core integer instructions are grouped into three broad functional classes.

Data Processing

- arithmetic
- logic
- comparison

add, sub
and, cmp

Memory Access

- loads
- stores
- transfer widths

ldr, str
ldp, stp

Control Flow

- branches
- calls
- returns

b, b.eq
bl, ret

Class 1: Data Processing

Data-processing instructions perform computation on register values.

Instruction	Operation	Effect
<code>add x3, x1, x2</code>	Addition	$x3 \leftarrow x1 + x2$
<code>sub x3, x1, x2</code>	Subtraction	$x3 \leftarrow x1 - x2$
<code>and x0, x1, x2</code>	Bitwise AND	$x0 \leftarrow x1 \& x2$
<code>orr x0, x1, x2</code>	Bitwise OR	$x0 \leftarrow x1 x2$

Key Characteristic

The operands are registers; these instructions do not directly read or write a memory operand.

Class 2: Memory Access

Memory instructions transfer data between registers and memory.

```
ldr x1, [x2]           // load 64 bits from memory into x1
```

```
str x5, [x3]           // store 64 bits from x5 to memory
```

```
ldr x1, [x2, #8]       // base address x2 plus byte offset 8
```

Bracket Notation

[x2] means: access memory using the address contained in x2.

Class 3: Control Flow

Control-flow instructions change which instruction executes next.

Unconditional

- `b target`: branch
- `bl function`: call
- `ret`: return

Conditional

- `b.eq target`: branch if equal
- `b.ne target`: branch if not equal
- Conditional branches may test NZCV flags.

Comparison and Branching

Common Pattern

```
cmp x0, x1  
b.eq equal
```

- `cmp` compares two register values.
- It updates condition flags rather than storing a data result.
- `b.eq` uses those flags to decide whether to branch.

How the Instruction Classes Work Together

Consider:

$$*p = *p + 5$$

where x0 contains the address represented by p.

Instruction	Class	Purpose
ldr x1, [x0]	Memory Access	Move the value from memory into a register
add x1, x1, #5	Data Processing	Compute the new value
str x1, [x0]	Memory Access	Write the result back to memory

RISC Connection

The sequence separates memory transfer from register computation.

Instruction Classification Exercise

Classify each instruction as **Data Processing**, **Memory Access**, or **Control Flow**.

Instruction	Your Classification
<code>add x3, x1, x2</code>	
<code>ldr x0, [x4]</code>	
<code>str w5, [x6]</code>	
<code>cmp x1, x2</code>	
<code>b.eq target</code>	
<code>bl function</code>	

Section 05

Summary & Self-Test

Topic 2 Summary

- **RISC Design:** regular formats, register-based computation, and explicit load/store memory access.
- **AArch64:** ARM's 64-bit architectural execution state.
- **A64:** fixed-width 32-bit instruction encoding used in AArch64 state.
- **Programmer-Visible State:** x0–x30, sp, xzr/wzr, PC state, and NZCV.
- **Instruction Classes:** Data Processing, Memory Access, and Control Flow.

Self-Test 1/3

- ① What architectural characteristics are commonly associated with RISC designs?
- ② What does it mean to say that AArch64 is a load/store architecture?
- ③ Why must a value normally be loaded into a register before an AArch64 arithmetic instruction can operate on it?
- ④ What is the distinction between **AArch64** and **A64**?

Self-Test 2/3

- 5 What happens to bits [63:32] of `x5` after an instruction writes to `w5`?
- 6 How general-purpose registers are visible in AArch64?
- 7 What are the roles of `xzr`, `sp`, and the program counter?
- 8 What do the N, Z, C, and V condition flags represent?

Self-Test 3/3

- 9 What are the three core instruction classes used in this course?
- 10 Classify each instruction: `add`, `ldr`, `str`, `cmp`, and `b.eq`.

Challenge

Explain how the sequence `ldr`, `add`, `str` demonstrates the RISC load/store model.

Wrap-Up and Next Steps

Topic 3: Machine-Level Data Representation

The next lecture examines how bit patterns manipulated by AArch64 instructions represent numerical values in registers and memory.

- Binary and hexadecimal representation
- Signed and unsigned integers
- Two's-complement representation
- Endianness and data alignment

Topic 2 established **what the AArch64 programmer can see**; Topic 3 examines **how data inside that visible state is represented**.